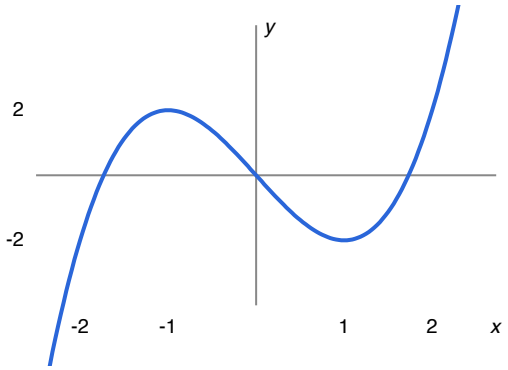
Graphs 8 — Gradient of a curve · Translating, reflecting and stretching graphs

Calculator allowed; graph paper for question 2. Marks in brackets. ★ = harder question. Key: blue = $y = \mathrm{f}(x)$, red = first transformed graph, green = second.

1 The graph of $y = x^3 - 3x$ is shown. **a** By drawing tangents, estimate the gradient of the curve at **i** $x = 0$ **ii** $x = 1.5$. **b** Where is the gradient zero? [6]



2 The temperature t °C of a cup of tea, m minutes after it is poured, is modelled by $t = \dfrac{900}{m + 15}$ for $0 \le m \le 30$. **a** Copy and complete the table (nearest degree). **b** Draw the graph of t against m. **c** Find the rate at which the tea is cooling, in °C per minute, when $m = 5$. [5]

m (min)	0	1	3	5	10	15	20	30
t (°C)	60			45		30		20

3 Here is a sketch of $y = \mathrm{f}(x)$ where $\mathrm{f}(x) = x^3$. **a** Sketch **i** $y = \mathrm{f}(x) + 2$ **ii** $y = \mathrm{f}(x - 2)$. **b** Write down where the point $(0, 0)$ is mapped to on each graph. [4]

4 The graph of $y = \mathrm{f}(x)$, where $\mathrm{f}(x) = (x - 1)^2 - 1$, has vertex $(1, -1)$ and crosses the x-axis at 0 and 2. Sketch **a** $y = 2\mathrm{f}(x)$ **b** $y = \mathrm{f}(2x)$, marking the vertex and intercepts of each. [4]

5 The diagram shows $y = \mathrm{f}(x)$ with $\mathrm{f}(x) = (x + 1)^2 - 4$ (vertex $(-1, -4)$, x-intercepts -3 and 1). **a** On the same axes sketch **i** $y = -\mathrm{f}(x)$ **ii** $y = \mathrm{f}(-x)$. **b** Describe the transformation from $y = \mathrm{f}(x)$ to $y = -\mathrm{f}(x)$. **c** Describe the transformation from $y = \mathrm{f}(x)$ to $y = \mathrm{f}(-x)$. [6]

Total: 25 marks

Check: gradient at a point = gradient of the tangent (rise ÷ run, with sign) · $\mathbf{f}(x) + a$ up, $\mathbf{f}(x + a)$ LEFT, $\mathbf{f}(x - a)$ right · $-\mathbf{f}(x)$ reflect in x-axis, $\mathbf{f}(-x)$ reflect in y-axis · $k\mathbf{f}(x)$ stretch y by k; $\mathbf{f}(kx)$ stretch x by $\frac{1}{k}$ · always say which axis and the scale factor or vector.